

Qualification Pack



Technician Grid Substation - Operation & Maintenance (66/11 KV, 33/11 KV)

QP Code: PSS/Q2001

Version: 1.0

NSQF Level: 4

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi-110021 || email:akbhagatin@yahoo.co.in



Qualification Pack

Contents

PSS/Q2001: Technician Grid Substation - Operation & Maintenance (66/11 KV, 33/11 KV)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
PSS/N1336: Working effectively with others	5
PSS/N2001: Use basic health and safety practices for power related work	9
PSS/N2002: Inspection, testing, operation and maintenance of substation transformers	17
PSS/N2003: Inspection, testing, operation and maintenance of Substation switchgear equipment ...	25
PSS/N2004: Manage protection-control-system and accessories in grid station	32
Assessment Guidelines and Weightage	39
<i>Assessment Guidelines</i>	39
<i>Assessment Weightage</i>	40
Acronyms	41
Glossary	42



Qualification Pack

PSS/Q2001: Technician Grid Substation - Operation & Maintenance (66/11 KV, 33/11 KV)

Brief Job Description

Technician Grid Substation Operation & Maintenance maintains all equipments in good operating condition. He takes up the schedule testing, operation, repair and maintenance of power transformer, switch gears, circuit breakers, isolators, power and control cable, over head conductor, cabling etc. under the supervision of Jr.Engineer (Executive).

Personal Attributes

Technician should be alert to observe minute details at the work place. The job holder should be able to cope with difficult work environment (such as heat, noise, height etc.). He/She must also have physical ability and stamina to stand for long working hours.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1336: Working effectively with others](#)
2. [PSS/N2001: Use basic health and safety practices for power related work](#)
3. [PSS/N2002: Inspection, testing, operation and maintenance of substation transformers](#)
4. [PSS/N2003: Inspection, testing, operation and maintenance of Substation switchgear equipment](#)
5. [PSS/N2004: Manage protection-control-system and accessories in grid station](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Technician
Country	India
NSQF Level	4
Credits	NA



Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NCO-2015/3113
Minimum Educational Qualification & Experience	I.T.I (Electrician trade) OR 10th Class with 2-3 Years of experience Junior Technician/Helper/Apprentice (Mandatory for 10th class)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	17/05/2017
Next Review Date	31/03/2022
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03485
NQR Version	1.0



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



Qualification Pack

- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



Qualification Pack

PSS/N2001: Use basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace
- PC5.** follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),
- PC6.** follow warning signs (danger, out of service, etc.) while working with electrical systems
- PC7.** use standard safe working practices when working at heights, confined areas and trenches
- PC8.** test any electrical equipment and system using insulated testing devices before touching them
- PC9.** ensure positive isolation of electrical equipment & system as per given standards
- PC10.** recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed
- PC11.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC12.** state methods of accident prevention in the work environment of the job role
- PC13.** state location of general health and safety equipment in the workplace
- PC14.** inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder
- PC15.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa
- PC16.** inspect grid station and its equipment routinely for any signs of oil and water leakage
- PC17.** store flammable materials and machine lubricating oil safely and correctly
- PC18.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC19.** ensure proper working condition of battery and battery charger
- PC20.** maintain electrolyte level of each cell with distilled water
- PC21.** maintain proper ventilation in battery room
- PC22.** apply good housekeeping practices at all times
- PC23.** identify common hazard signs displayed in various areas



Qualification Pack

PC24. retrieve and/or point out documents that refer to health and safety in the workplace

PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly

Fire Safety

To be competent, the user/individual on the job must be able to:

PC26. use the various appropriate fire extinguishers on different types of fires correctly

PC27. distinguish between various types of fire

PC28. demonstrate rescue techniques applied during fire hazard

PC29. demonstrate good housekeeping in order to prevent fire hazards

PC30. demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC31. demonstrate how to free a person from electrocution

PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC33. demonstrate basic techniques of bandaging

PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC37. demonstrate the artificial respiration and the cpr process

PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work

PC39. write accident/incident report or dictate a report to another person, and send report to person responsible

PC40. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. escalation matrix and procedures for reporting work and employment related issues

KU5. meaning of hazards and risks

KU6. health and safety hazards commonly present in the work environment and related precautions

KU7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible



Qualification Pack

- KU8.** possible causes of risk and accident
- KU9.** methods of accident prevention
- KU10.** safe working practices when working with tools and machines
- KU11.** safe working practices while working at various hazardous sites
- KU12.** where to find all the general health and safety equipment in the workplace
- KU13.** various dangers associated with the use of electrical equipment
- KU14.** positive isolation of electrical equipment and system
- KU15.** safe handling and disposal of hazardous power plant wastes
- KU16.** use of emission and pollution control devices and measures taken to control pollution
- KU17.** various safety procedures and equipment used to work at heights, trenches and confined places
- KU18.** safe working practices specific to working with electrical equipment & system e.g. lock out/tag out, ptw, etc.
- KU19.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU20.** importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
- KU21.** precautionary activities taken to prevent fire accident
- KU22.** various causes of fire
- KU23.** techniques of using the different fire extinguishers
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire
- KU26.** emergency rescue techniques applied during a fire hazard
- KU27.** various types of safety signs and what they mean
- KU28.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge.
- GS2.** note down observations (if any) related to the operation/maintenance.
- GS3.** read and interpret the process required for different types of manuals for maintenance.
- GS4.** read and interpret the flowchart of all parts of an assembly.
- GS5.** read manuals and documents to understand the product-details & how they can be used.
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor.
- GS7.** effectively communicate with the team members.
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor.
- GS9.** communicate clearly with the colleague on the issues faced during query/fault.
- GS10.** follow colleague/contractor rule-based decision making process.
- GS11.** take decisions with systematic course of actions and/or response.



Qualification Pack

- GS12.** planning and organization of tasks to meet deadlines.
- GS13.** build customer relationships and use customer centric approach.
- GS14.** seek and comprehend operation related inputs for clarification
- GS15.** find ways of modifying difficult operating stages to make it operation friendly
- GS16.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results.
- GS17.** quick approach and solution towards faults repairing.
- GS18.** critically evaluate operation parameters in relation to system normality
- GS19.** develop a holistic and comprehensive profile of grid station on segregated discrete process.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	11	46	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	-	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	-	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	-	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace	1	1	-	-
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),	1	2	-	-
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems	1	1	-	-
PC7. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC8. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC9. ensure positive isolation of electrical equipment & system as per given standards	1	2	-	-
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed	1	2	-	-
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	1	-	-
PC12. state methods of accident prevention in the work environment of the job role	-	2	-	-
PC13. state location of general health and safety equipment in the workplace	-	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder	-	2	-	-
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa	-	2	-	-
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage	-	2	-	-
PC17. store flammable materials and machine lubricating oil safely and correctly	-	2	-	-
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	1	-	-
PC19. ensure proper working condition of battery and battery charger	1	1	-	-
PC20. maintain electrolyte level of each cell with distilled water	-	2	-	-
PC21. maintain proper ventilation in battery room	-	1	-	-
PC22. apply good housekeeping practices at all times	1	2	-	-
PC23. identify common hazard signs displayed in various areas	-	2	-	-
PC24. retrieve and/or point out documents that refer to health and safety in the workplace	-	2	-	-
PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly	-	3	-	-
Fire Safety	4	9	-	-
PC26. use the various appropriate fire extinguishers on different types of fires correctly	1	1	-	-
PC27. distinguish between various types of fire	1	2	-	-
PC28. demonstrate rescue techniques applied during fire hazard	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate good housekeeping in order to prevent fire hazards	-	2	-	-
PC30. demonstrate the correct use of a fire extinguisher	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	21	-	-
PC31. demonstrate how to free a person from electrocution	1	2	-	-
PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	-	3	-	-
PC33. demonstrate basic techniques of bandaging	1	2	-	-
PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC37. demonstrate the artificial respiration and the cpr process	1	2	-	-
PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC39. write accident/incident report or dictate a report to another person, and send report to person responsible	1	2	-	-
PC40. demonstrate correct method to move injured people and others during an emergency	1	2	-	-
NOS Total	24	76	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2001
NOS Name	Use basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022



Qualification Pack

PSS/N2002: Inspection, testing, operation and maintenance of substation transformers

Description

The Grid Station Technician (O&M) inspects all the equipment installed in the switchyard as well as in control room of a Grid Station. This also covers daily, monthly, quarterly and annual inspection and maintenance schedule, including preventive maintenance and breakdown maintenance of transformers as per approved inspection schedule.

Scope

This unit/task covers the following:

- types of maintenance: preventive, breakdown, predictive maintenance and condition monitoring
- operation, inspection, testing, repair and maintenance of transformers installed in 66kV or 33kV Grid Station.
- job specific safety

Elements and Performance Criteria

Types of maintenance: preventive, breakdown, predictive maintenance and condition monitoring

To be competent, the user/individual on the job must be able to:

- PC1.** be aware of the grid station infrastructure, number of bays, number of incoming and outgoing feeders, load management through single or double bus, number of power transformer, station transformer, control panel, battery panels, area network, yard lighting etc.
- PC2.** prepare the detail for inspection and maintenance of substation equipment as per approved schedule
- PC3.** arrange shutdown for planned maintenance
- PC4.** carry out preventive maintenance and breakdown maintenance of all equipment
- PC5.** prepare and maintain all records of tests, results, repairs of all equipment
- PC6.** ensure store of all testing instruments, tools & tackles in proper healthy state, safe and for easy access
- PC7.** maintain inventories of spare parts for all equipments
- PC8.** perform routine operation, maintenance and trouble shooting of all substation equipments
- PC9.** identify faulty equipment and safe isolation without disturbing of other equipment
- PC10.** ensure safety chart, first aid box, switchgear handles, fire extinguishers, PPEs and discharge rod are placed at proper location
- PC11.** apply CEA, SERC regulations and IE Act 2003

Inspection, testing and maintenance of transformers installed in Grid Station

To be competent, the user/individual on the job must be able to:

- PC12.** check general appearance of bushings and oil level of a transformer (power and station), cleaning of HV & LV bushings



Qualification Pack

- PC13.** manage auxiliary cooling systems like cooling fans, pumps, oil coolers in the different types of transformers
- PC14.** demonstrate knowledge of OLTC and its mechanism in a transformer
- PC15.** check various gauges like oil level gauge, pressure vacuum gauge, oil temperature gauge, winding temperature gauge etc.
- PC16.** check air release in main tanks, bucholtz relays
- PC17.** neutralize earth connections at both ends prior to after conduct testing main operation
- PC18.** set alarm circuits and trip circuit of transformer
- PC19.** ensure working of fire protection system
- PC20.** ensure working of transformer
- PC21.** check and repair the oil level and leakage CTs, PTs and CVTs
- PC22.** check general appearance of bushings, terminal block and earth connection of secondary circuit

Job specific safety

To be competent, the user/individual on the job must be able to:

- PC23.** adopt guidelines, procedures and safety regulations laid down in the organization for safety
- PC24.** use personal protective equipment (PPE) for proper safety
- PC25.** ensure work area is clean and safe from hazards
- PC26.** use safety tags or other warning boards before commencing work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** how to follow relevant standards, working procedures and policies of organization
- KU2.** main purpose and object of organization
- KU3.** department structure
- KU4.** reporting structure
- KU5.** conditions and terms of own employment
- KU6.** own job role and responsibilities
- KU7.** sources of information
- KU8.** knowledge of work area
- KU9.** working safely
- KU10.** cleanness of working area, maintain 5S
- KU11.** principles of electricity
- KU12.** power system: flow of power, generation, transmission and distribution, number of bays, number of incoming and outgoing feeders, load management through single or double bus, substation network, ring system, back feed etc.
- KU13.** operating principle of power transformer, its components, auxiliary components and accessories
- KU14.** instrument transformers like CTs, PTs, and CTV's



Qualification Pack

- KU15.** key faults that occur in major equipments like power transformer, distribution transformer, circuit breakers, control and relay panels etc. their causes and protection system to avoid break down.
- KU16.** operating function of transformer, oil filter machine, inlet & outlet connection, complete tools, tackles and safety gadgets required in grid station
- KU17.** use of safety, T&P and testing equipment like multimeter, partial discharging kit, circuit breakers testing kit, meggar, sdf6 gas leakage kit, digital tong tester, oil testing kit, battery testing kit, hydraulic crimping kit high voltage testing kit, protection testing kit, fault locator kit, tool box kit and safety gadgets like discharge rod, earth chain, neon tester, signage, danger notice, carbon tape etc.
- KU18.** approved maintenance procedures and regulation
- KU19.** safety precautions as per safety manual
- KU20.** to keep records of all equipment like name plate, pre- commission test report and manuals
- KU21.** the importance of reporting problem to junior engineer, officer incharge
- KU22.** calibration details/schedule of testing and periodic maintenance of equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively in writing as per requirement of site work
- GS2.** write the information communicated by the engineer or in-charge of work
- GS3.** write properly about the technical problems and other conditions of site
- GS4.** note down of observations, critical points and location of site related work
- GS5.** reading, understanding of written sentences and paragraphs in work related documents
- GS6.** writing and using metric system for all measurements
- GS7.** how to interpret the process required for performing of work
- GS8.** how to read, interpret and understand the rules and method stated in the documents
- GS9.** how to read equipment manuals and understand the equipment operation and process requirement
- GS10.** discuss task lists, schedules and activities with the engineer
- GS11.** effectively communicate with the team/group members
- GS12.** listen to the information given by the junior engineer
- GS13.** communicate clearly with the team and other staff
- GS14.** make work related judgements appropriately
- GS15.** identify complex problems and review related information to develop and evaluate
- GS16.** follow organization rule based decision making process
- GS17.** take decisions with systematic course of actions and/or response
- GS18.** plan and organize tasks to meet deadlines
- GS19.** plan in advance the maintenance schedule- daily, weekly and monthly.
- GS20.** organize manpower, tools and kits for testing and maintenance.
- GS21.** plan and organize the shutdown of equipment



Qualification Pack

- GS22.** plan, if any, special tool and testing kit required for maintenance
- GS23.** plan work of maintenance completed within specified time
- GS24.** plan return to shutdown permit in time
- GS25.** build customer relationships and use customer centric approach
- GS26.** identify problems and review related information to develop and evaluate options and implement solutions
- GS27.** prioritize and plan for solving problem
- GS28.** take help from the junior engineer to solve the problems
- GS29.** monitor solving problems and to take corrective action with individuals and organizations
- GS30.** analyse problems and changes in conditions, operations, and the environment to solve problems
- GS31.** analyze the problem seen in the equipment
- GS32.** collect the information and technical data and define process for doing testing and maintenance
- GS33.** critically evaluate operation parameters in relation to power transformer features intended
- GS34.** develop a holistic and comprehensive profile of power transformer based on segregated discrete process stages of blank forming processes



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Types of maintenance: preventive, breakdown, predictive maintenance and condition monitoring</i>	14	28	-	-
PC1. be aware of the grid station infrastructure, number of bays, number of incoming and outgoing feeders, load management through single or double bus, number of power transformer, station transformer, control panel, battery panels, area network, yard lighting etc.	2	2	-	-
PC2. prepare the detail for inspection and maintenance of substation equipment as per approved schedule	2	3	-	-
PC3. arrange shutdown for planned maintenance	2	2	-	-
PC4. carry out preventive maintenance and breakdown maintenance of all equipment	2	2	-	-
PC5. prepare and maintain all records of tests, results, repairs of all equipment	2	2	-	-
PC6. ensure store of all testing instruments, tools & tackles in proper healthy state, safe and for easy access	-	3	-	-
PC7. maintain inventories of spare parts for all equipments	1	3	-	-
PC8. perform routine operation, maintenance and trouble shooting of all substation equipments	1	3	-	-
PC9. identify faulty equipment and safe isolation without disturbing of other equipment	1	3	-	-
PC10. ensure safety chart, first aid box, switchgear handles, fire extinguishers, PPEs and discharge rod are placed at proper location	-	3	-	-
PC11. apply CEA, SERC regulations and IE Act 2003	1	2	-	-
<i>Inspection, testing and maintenance of transformers installed in Grid Station</i>	11	31	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check general appearance of bushings and oil level of a transformer (power and station), cleaning of HV & LV bushings	-	3	-	-
PC13. manage auxiliary cooling systems like cooling fans, pumps, oil coolers in the different types of transformers	-	3	-	-
PC14. demonstrate knowledge of OLTC and its mechanism in a transformer	2	3	-	-
PC15. check various gauges like oil level gauge, pressure vacuum gauge, oil temperature gauge, winding temperature gauge etc.	2	3	-	-
PC16. check air release in main tanks, bucholtz relays	2	2	-	-
PC17. neutralize earth connections at both ends prior to after conduct testing main operation	1	3	-	-
PC18. set alarm circuits and trip circuit of transformer	2	3	-	-
PC19. ensure working of fire protection system	-	3	-	-
PC20. ensure working of transformer	1	2	-	-
PC21. check and repair the oil level and leakage CTs, PTs and CVTs	-	3	-	-
PC22. check general appearance of bushings, terminal block and earth connection of secondary circuit	1	3	-	-
<i>Job specific safety</i>	3	13	-	-
PC23. adopt guidelines, procedures and safety regulations laid down in the organization for safety	1	3	-	-
PC24. use personal protective equipment (PPE) for proper safety	-	4	-	-
PC25. ensure work area is clean and safe from hazards	-	3	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. use safety tags or other warning boards before commencing work	2	3	-	-
NOS Total	28	72	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2002
NOS Name	Inspection, testing, operation and maintenance of substation transformers
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Operation & Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	17/05/2017
Next Review Date	31/03/2022
NSQC Clearance Date	NA



Qualification Pack

PSS/N2003: Inspection, testing, operation and maintenance of Substation switchgear equipment

Description

The Grid Substation Technician inspects all the equipment installed in the switchyard as well as in control room of a Grid Substation. This also covers daily, monthly, quarterly and annual inspection and maintenance schedule, including preventive maintenance and breakdown maintenance as per approved inspection schedule of the equipments.

Scope

This unit/task covers the following:

- types of maintenance: preventive, breakdown, predictive maintenance and condition monitoring
- inspection, testing, repair and maintenance of switch gears installed in 66kV or 33kV Grid Substation
- job specific safety

Elements and Performance Criteria

Types of maintenance: preventive, breakdown, predictive maintenance and condition monitoring

To be competent, the user/individual on the job must be able to:

- PC1.** work out on grid substation infrastructure, number of bays, number of incoming and outgoing feeders, load management through single or double bus, number of power transformers, station transformers, control panel, battery panels, area network, yard lighting etc.
- PC2.** perform inspection and maintenance of substation equipment as per approved schedule
- PC3.** carry out shutdown for planned maintenance
- PC4.** carry out preventive maintenance and breakdown maintenance of all equipment
- PC5.** maintain records of tests, results, repairs, and maintenance of all equipment
- PC6.** maintain storing of all testing instruments, tools & tackles in proper healthy state, safe and easily available
- PC7.** maintain inventories of spare parts for all equipments
- PC8.** perform routine operation, maintenance and trouble shooting of all substation equipments
- PC9.** identify faulty equipment and safe isolation without disturbing of other equipment
- PC10.** maintain job cards of each equipment
- PC11.** ensure safety chart, first aid box, switchgear handles, fire extinguishers, PPEs and discharge rod are placed at proper location.
- PC12.** apply basic knowledge of CEA, SERC regulations and IE Act 2003

Inspection, testing, repair and maintenance of switchgears installed in Grid Station

To be competent, the user/individual on the job must be able to:

- PC13.** inspect the circuit breaker as per manual/recommendation of manufacturer
- PC14.** check tripping mechanism for short circuit, over current and earth fault
- PC15.** check signage/flag status of trip due to fault, or operation for ON/OFF and earth fault



Qualification Pack

- PC16.** ensure operating handles are in healthy state and placed at appropriate location
- PC17.** check loose connection in control wiring
- PC18.** check breaker assembly frame, condition of foundation and foundation bolt
- PC19.** clean main contacts, earth blade and spring assembly
- PC20.** apply petroleum jelly to all contacts and lubricate all moving parts/assembly

Job specific safety

To be competent, the user/individual on the job must be able to:

- PC21.** adopt guidelines and procedures and safety regulations laid down in the organization for safety
- PC22.** use personal protective equipment (PPE) for proper safety
- PC23.** ensure work area is clean and safe from hazards.
- PC24.** use safety tags or other warning boards before commencing work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards, working procedures and policies of organization
- KU2.** main purpose and objective of organization
- KU3.** department structure
- KU4.** reporting structure
- KU5.** conditions and terms of own employment
- KU6.** own job role and responsibilities
- KU7.** sources of information
- KU8.** knowledge of work area
- KU9.** working safely
- KU10.** cleanness of working area, maintain 5s
- KU11.** interpersonal relations
- KU12.** principles of electricity
- KU13.** power system: flow of power, generation, transmission and distribution, number of bays, number of incoming and outgoing feeders, load management through single or double bus, substation network, ring system, back feed etc.
- KU14.** line components towers, poles, single circuit, double circuit, over head, underground conductors and cables
- KU15.** gantry structure, structure lay out, types of porcelain insulators, overhead conductors, clamps used in substation
- KU16.** operating principle of switch gears (CB), how it operates under fault current, benefits of operating medium of OCB, MOCB, ACB, SF, Vacuum circuit breakers
- KU17.** fault current due to short circuit, over current and earth fault, their impact and protection system
- KU18.** control panel and its in-built measuring instruments, accessories like heater, lamp, door switch, HRC fuse, relays, auxiliary, ICTs etc.



Qualification Pack

- KU19.** battery panel, trickle charging, battery status, electrolyte level, specific gravity of electrolyte, safety measures in repair and maintenance, ventilation etc.
- KU20.** tripping mechanism, short circuit, earth fault, over current, low frequency etc.
- KU21.** key faults that occur in major equipment like power transformer, distribution transformer, circuit breakers, control and relay panels etc. their causes and protection system to avoid break down.
- KU22.** lightning arrestors (LA) functioning
- KU23.** types of earthing used in grid station, its significance, why earth connection with each equipments, measurement of earth resistance
- KU24.** shunt capacitor bank, its function to improve pf, switching operation, repair, replacement and maintenance
- KU25.** complete tools, tackles and safety gadgets required in grid station
- KU26.** use of safety, T&P and testing equipment like multimeter, partial discharging kit, circuit breakers testing kit, megger, SDF6 gas leakage kit, digital tong tester, oil testing kit, battery testing kit, hydraulic crimping kit high voltage testing kit, protection testing kit, fault locator kit, tool box kit and safety gadgets like discharge rod, earth chain, neon tester, signage, danger notice, carbon tape etc.
- KU27.** approved maintenance procedures and regulation
- KU28.** how to take safety precautions as per safety manual
- KU29.** how to keep records of all equipment like name plate, pre-commission test report and manuals
- KU30.** the importance of reporting problem to junior engineer, officer incharge

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively in writing as per requirement of site work
- GS2.** write the information communicated by the engineer or in-charge of work
- GS3.** write properly about the technical problems and other conditions of site
- GS4.** note down of observations, critical points and location of site related work
- GS5.** reading, written sentences and paragraphs in work related documents
- GS6.** writing and using metric system for all measurements
- GS7.** how interpret the process required for the performing of work
- GS8.** how to read, interpret and understand the rules and method stated in the documents
- GS9.** how to read equipment manuals and understand the equipment operation and process requirement
- GS10.** discuss task lists, schedules and activities with the engineer
- GS11.** effectively communicate with the team/group members
- GS12.** listen the information given by the junior engineer
- GS13.** communicate clearly with the team and other staff
- GS14.** make work related judgements appropriately
- GS15.** identify complex problems and review related information to develop and evaluate



Qualification Pack

- GS16.** follow organization rule based decision making process
- GS17.** take decisions with systematic course of actions and/or response plan and
- GS18.** plan and organize tasks to meet deadlines
- GS19.** plan in advance the maintenance schedule- daily, weekly and monthly.
- GS20.** properly organize manpower and tools and kits for testing and maintenance.
- GS21.** plan and organize the shutdown of equipment
- GS22.** plan, if any, special tool and testing kit required for maintenance
- GS23.** plan work of maintenance completed within specified time
- GS24.** plan return to shutdown permit in time
- GS25.** build customer relationships and use customer centric approach
- GS26.** identify problems and review related information to develop and evaluate options and implement solutions
- GS27.** prioritize and plan for solving problem
- GS28.** take help from the junior engineer to solve the problems
- GS29.** monitor solving problems and to take corrective action with individuals and organizations
- GS30.** analyse problems and changes in conditions, operations, and the environment to solve problems
- GS31.** analyze the problem seen in the equipment
- GS32.** collect the information and technical data and define process for doing testing and maintenance
- GS33.** critically evaluate operation parameters in relation to switchgear features intended
- GS34.** develop a holistic and comprehensive profile of switchgear based on segregated discrete process stages of blank forming processes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Types of maintenance: preventive, breakdown, predictive maintenance and condition monitoring</i>	17	35	-	-
PC1. work out on grid substation infrastructure, number of bays, number of incoming and outgoing feeders, load management through single or double bus, number of power transformers, station transformers, control panel, battery panels, area network, yard lighting etc.	4	6	-	-
PC2. perform inspection and maintenance of substation equipment as per approved schedule	2	3	-	-
PC3. carryout shutdown for planned maintenance	2	2	-	-
PC4. carry preventive maintenance and breakdown maintenance of all equipment	2	2	-	-
PC5. maintain records of tests, results, repairs, and maintenance of all equipment	2	2	-	-
PC6. maintain storing of all testing instruments, tools & tackles in proper healthy state, safe and easily available	-	3	-	-
PC7. maintain inventories of spare parts for all equipments	1	3	-	-
PC8. perform routine operation, maintenance and trouble shooting of all substation equipments	1	3	-	-
PC9. identify faulty equipment and safe isolation without disturbing of other equipment	1	3	-	-
PC10. maintain job cards of each equipment	-	3	-	-
PC11. ensure safety chart, first aid box, switchgear handles, fire extinguishers, pps and discharge rod are placed at proper location.	-	3	-	-
PC12. apply basic knowledge of CEA, SERC regulations and IE Act 2003	2	2	-	-
<i>Inspection, testing repair and maintenance of switchgears installed in Grid Station</i>	8	24	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. inspect the circuit braker as per manual/recommendation of manufacturer	2	2	-	-
PC14. check tripping mechanism for short circuit, over current and earth fault	2	3	-	-
PC15. check signage/flag status of trip due to fault, or operation for ON/OFF and earth fault	-	3	-	-
PC16. ensure operating handles are in healthy state and placed at appropriate location	-	3	-	-
PC17. check loose connection in control wiring	2	3	-	-
PC18. check breaker assembly frame, condition of foundation and foundation bolt	1	3	-	-
PC19. clean main contacts, earth blade and spring assembly	1	3	-	-
PC20. apply petroleum jelly to all contacts and lubricate all moving parts/assembly	-	4	-	-
<i>Job specific safety</i>	4	12	-	-
PC21. adopt guidelines and procedures and safety regulations laid down in the organization for safety	2	3	-	-
PC22. use personal protective equipment (PPE) for proper safety	-	3	-	-
PC23. ensure work area is clean and safe from hazards.	-	3	-	-
PC24. use safety tags or other warning boards before commencing work	2	3	-	-
NOS Total	29	71	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2003
NOS Name	Inspection, testing, operation and maintenance of Substation switchgear equipment
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Operation & Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	17/05/2017
Next Review Date	31/03/2022
NSQC Clearance Date	NA



Qualification Pack

PSS/N2004: Manage protection-control-system and accessories in grid station

Description

The Grid Substation Technician inspects all the equipment installed in the switchyard as well as in the control room of a Grid Substation. This also covers daily, monthly, quarterly and annual inspection and maintenance schedule, including preventive maintenance and breakdown maintenance as per approved inspection schedule of equipment.

Scope

This unit/task covers the following:

- job specific safety
- maintenance of battery bank and shunt capacitor bank: preventive, breakdown and condition monitoring
- inspection, testing, repair and maintenance of earthing, LA's, control panels and switch yard area of 66kV or 33kV Grid Station.

Elements and Performance Criteria

Job specific safety

To be competent, the user/individual on the job must be able to:

- PC1.** adopt guidelines, procedures and safety regulations laid down in the organization for safety
- PC2.** use personal protective equipment (PPE) for proper safety
- PC3.** ensure work area is clean and safe from hazards
- PC4.** use safety tags or other warning boards before commencing work

Maintenance of Battery bank and Shunt capacitor bank: preventive, breakdown and condition monitoring

To be competent, the user/individual on the job must be able to:

- PC5.** the condition of battery and battery charger
- PC6.** that the float charge is always in on condition
- PC7.** to keep exhaust fan in working condition and maintain proper ventilation
- PC8.** about sufficient air inlet provision in the battery room
- PC9.** to maintain electrolyte level of each cell with distilled water
- PC10.** the functioning of capacitor bank and shunt capacitor bank for improvement of the power factor

Inspection, testing, repair and maintenance of earthing, LAs, switch yard area and control panels of 66kV or 33kV Grid Station

To be competent, the user/individual on the job must be able to:

- PC11.** ensure earth strips are intact and firmly connected in each gantry and equipment
- PC12.** manage earth resistance of each earth pit marked with date and next due date
- PC13.** pour water into each earth pit during summer (low ground water level)
- PC14.** provide double earth connection to each equipment



Qualification Pack

- PC15.** clean the lightning arrestors (LA) in stacks
- PC16.** check surge counter and leakage current in la
- PC17.** check of hot spots by thermo-vision camera in switch yard area
- PC18.** check overhead conductor in the switch yard area
- PC19.** clean of bush bar string insulators
- PC20.** tighten the t clamps and pg clamps
- PC21.** clean the bus bar
- PC22.** ensure functioning of fire extinguishes regularly or as defined in the guidelines
- PC23.** have switchyard illumination, luminaires and replacement of fused bulbs replaced
- PC24.** ensure painting of all equipment
- PC25.** check status of relays O/C & E/F in the control panel in their settings, flag etc.
- PC26.** check status of indicators, meters (Volt, Ampere, Watt, PF, HZ, Eergy etc.)
- PC27.** manage and test terminal block for availability of secondary current and voltage
- PC28.** check status of door switch, lamp and heater in side the chamber
- PC29.** mark the door of control panel with name of feeder, number the same, ensure the door is approachable, clean, dust free and vermin protected
- PC30.** ensure the cable entry hole beneath the control panel should be properly plugged to avoid raptile entry and damage to wiring, instruments, accessories
- PC31.** check terminal connection block control, cables are connected in proper sequence with color code and ferrule
- PC32.** ensure the control wires are layed in proper bunch in the concealed trays
- PC33.** check status of hrc fuse (PT and circuit) for correct rating
- PC34.** mount the ICT, auxiliary relays and other accessories in healthy and proper mannar

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** follow relevant standards, working procedures and policies of organization
- KU2.** main purpose and object of organization
- KU3.** department structure
- KU4.** reporting structure
- KU5.** conditions and terms of own employment
- KU6.** own job role and responsibilities
- KU7.** sources of information
- KU8.** knowledge of work area
- KU9.** working safely
- KU10.** cleanness of working area, maintain 5s
- KU11.** interpersonal relations
- KU12.** principles of electricity
- KU13.** fault current due to short circuit, over current and earth fault, their impact and protection system function



Qualification Pack

- KU14.** tripping mechanism like short circuit, earth fault, over current, low frequency etc.
- KU15.** occurrence of key faults in major equipment like power transformer, distribution transformer, circuit breakers, control and relay panels etc. their causes and protection system to avoid breakdown
- KU16.** control panel and its in built measuring instruments, accessories like heater, lamp, door switch, HRC fuse, Relays, Auxiliary, ICTs etc.
- KU17.** battery panel, trickle charging, battery status, electrolyte level, specific gravity of electrolyte, safety measures in repair and maintenance, ventilation etc.
- KU18.** functioning of lightning arrestors (LA)
- KU19.** types of earthing used in grid substation, its significance, earth connection with each equipments, measurement of earth resistance
- KU20.** functionality of shunt capacitor bank, how to improve pf, switching operation, repair, replacement and maintenance
- KU21.** all tools, tackles and safety gadgets required in grid substation
- KU22.** use of safety, T&P and testing equipment like multimeter, partial discharging kit, circuit breakers testing kit, meggar, SDF6 gas leakage kit, digital tong tester, oil testing kit, battery testing kit, hydraulic crimping kit high voltage testing kit, protection testing kit, fault locator kit, tool box kit and safety gadgets like discharge rod, earth chain, neon tester, signage, danger notice, carbon tape etc.
- KU23.** working within approved maintenance procedures and regulation
- KU24.** safety precautions as per safety manual
- KU25.** importance of keeping records of all equipment like name plate, pre- commission test report and manuals
- KU26.** importance of reporting problem to junior engineer, officer incharge

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively in writing as per requirement of site work
- GS2.** write the information communicated by the engineer or in-charge of work
- GS3.** write properly about the technical problems and other conditions of site
- GS4.** note down of observations, critical points and location of site related work
- GS5.** reading, written sentences and paragraphs in work related documents
- GS6.** writing and using metric system for all measurements
- GS7.** how to interpret the process required for performing of work
- GS8.** how to read, interpret and understand the rules and method stated in the documents
- GS9.** how to read equipment manuals and understand the equipment operation and process requirement
- GS10.** discuss task lists, schedules and activities with the engineer
- GS11.** effectively communicate with the team/group members
- GS12.** listen the information given by the junior engineer
- GS13.** communicate clearly with the team and other staff



Qualification Pack

- GS14.** make work related judgements appropriately
- GS15.** identify complex problems and review related information to develop and evaluate
- GS16.** follow organization rule based decision making process
- GS17.** take decisions with systematic course of actions and/or response
- GS18.** plan and organize tasks to meet deadlines
- GS19.** plan in advance maintenance schedule daily, weekly and monthly
- GS20.** organize manpower, tools and kits for testing and maintenance
- GS21.** plan and organize the shutdown of equipment
- GS22.** plan for special tool and testing kit required for maintenance
- GS23.** complete maintenance work within specified time
- GS24.** return from shutdown in the permitted time
- GS25.** build customer relationships and use customer centric approach
- GS26.** identify problems and review related information to develop and evaluate options and implement solutions
- GS27.** prioritize and plan for solving problem
- GS28.** take help from the junior engineer to solve the problems
- GS29.** monitor solving problems and take corrective action with individuals and organizations
- GS30.** analyse problems and changes in conditions, operations, and the environment to solve problems
- GS31.** analyze the problem seen in the equipment
- GS32.** collect the information and technical data and define process for doing testing and maintenance
- GS33.** critically evaluate operation parameters in relation to dc control features intended
- GS34.** develop a holistic and comprehensive profile of control system based on segregated discrete process stages of blank forming processes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Job specific safety</i>	3	8	-	-
PC1. adopt guidelines, procedures and safety regulations laid down in the organization for safety	2	2	-	-
PC2. use personal protective equipment (PPE) for proper safety	-	2	-	-
PC3. ensure work area is clean and safe from hazards	-	2	-	-
PC4. use safety tags or other warning boards before commencing work	1	2	-	-
<i>Maintenance of Battery bank and Shunt capacitor bank: preventive, breakdown and condition monitoring</i>	3	12	-	-
PC5. the condition of battery and battery charger	1	2	-	-
PC6. that the float charge is always in on condition	-	2	-	-
PC7. to keep exhaust fan in working condition and maintain proper ventilation	-	2	-	-
PC8. about sufficient air inlet provision in the battery room	-	2	-	-
PC9. to maintain electrolyte level of each cell with distilled water	2	2	-	-
PC10. the functioning of capacitor bank and shunt capacitor bank for improvement of the power factor	-	2	-	-
<i>Inspection, testing, repair and maintenance of earthing, LAs, switch yard area and control panels of 66kV or 33kV Grid Station</i>	16	58	-	-
PC11. ensure earth strips are intact and firmly connected in each gantry and equipment	1	3	-	-
PC12. manage earth resistance of each earth pit marked with date and next due date	-	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. pour water into each earth pit during summer (low ground water level)	1	2	-	-
PC14. provide double earth connection to each equipment	-	3	-	-
PC15. clean the lightning arrestors (LA) in stacks	1	3	-	-
PC16. check surge counter and leakage current in la	-	3	-	-
PC17. check of hot spots by thermo-vision camera in switch yard area	-	3	-	-
PC18. check overhead conductor in the switch yard area	-	3	-	-
PC19. clean of bush bar string insulators	-	3	-	-
PC20. tighten the t clamps and pg clamps	-	3	-	-
PC21. clean the bus bar	-	3	-	-
PC22. ensure functioning of fire extinguishes regularly or as defined in the guidelines	-	3	-	-
PC23. have switchyard illumination, luminaires and replacement of fused bulbs replaced	-	3	-	-
PC24. ensure painting of all equipment	2	2	-	-
PC25. check status of relays O/C & E/F in the control panel in their settings, flag etc.	2	2	-	-
PC26. check status of indicators, meters (Volt, Ampere, Watt, PF, HZ, Eergy etc.)	2	2	-	-
PC27. manage and test terminal block for availability of secondary current and voltage	1	2	-	-
PC28. check status of door switch, lamp and heater in side the chamber	-	3	-	-
PC29. mark the door of control panel with name of feeder, number the same, ensure the door is approachable, clean, dust free and vermin protected	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. ensure the cable entry hole beneath the control panel should be properly plugged to avoid raptile entry and damage to wiring, instruments, accessories	1	2	-	-
PC31. check terminal connection block control, cables are connected in proper sequence with color code and ferrule	1	2	-	-
PC32. ensure the control wires are layed in proper bunch in the concealed trays	1	1	-	-
PC33. check status of hrc fuse (PT and circuit) for correct rating	1	1	-	-
PC34. mount the ICT, auxiliary relays and other accessories in healthy and proper mannar	1	2	-	-
NOS Total	22	78	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2004
NOS Name	Manage protection-control-system and accessories in grid station
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Operation & Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	17/05/2017
Next Review Date	31/03/2022
NSQC Clearance Date	NA

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2001.Use basic health and safety practices for power related work	24	76	-	-	100	15
PSS/N2002.Inspection, testing, operation and maintenance of substation transformers	28	72	-	-	100	25
PSS/N2003.Inspection, testing, operation and maintenance of Substation switchgear equipment	29	71	-	-	100	25
PSS/N2004.Manage protection-control-system and accessories in grid station	22	78	-	-	100	30
Total	133	367	-	-	500	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
V	Volt



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.